

Unlike the late autumn of 1950, when the world's first electronic computer ENIAC had already been born for several years, computer science was gradually becoming an independent discipline, and computer technology was beginning to develop rapidly. As an emerging technological field, it had just received widespread attention and investment. The famous British scientist Alan Mathison Turing, in his seminal work "Computing Machinery and Intelligence", elaborated in detail on the thought experiment of the imitation game, proposed the standard of the Turing test, and discussed the possibility, limitations, and future directions of machine intelligence. He argued that if a machine could successfully impersonate a human in the imitation game, then the machine could be considered intelligent. These wise insights have flickered for 76 years, and the world has entered the spring of March 2026. The machines of yesteryear have transformed into computers and mobile phones that have entered thousands of households. The information industry has developed rapidly, the data ecosystem is changing with each passing day, and intelligent machines have sprung up in large numbers. Whether they are embodied intelligent robots, non-humanoid intelligent machines, smart equipment, intelligent accessories like smart glasses, or various smart terminals running artificial intelligence application services, they have all become commonplace. The latest craze is undoubtedly brought by the open-source artificial intelligence application system Openclaw, whose iconic symbol – a little lobster – has grabbed news and trending headlines. This article is based on Turing's work, but not simply as a tribute to the late Turing, nor to criticize or blame anyone. It merely aims to raise new developments and situations in the field of computing machinery and intelligence, and will quote some of his original statements, which I believe does not constitute infringement.

****1. Imitation Game****

I propose to consider the question: "Can machines do evil?" Or, in our Chinese way of saying it: "Can machines do bad things?" This should begin with defining the meaning of the judgment "do evil" and "do bad things". If the meaning of "do evil" and "do bad things" is to be discovered by examining how they are commonly used, it is easy to avoid concluding that this meaning and the answer to the question should be sought in statistical surveys like the Gallup poll in the West. Instead of attempting such a definition, it is better to replace this question with another one that is closely related to the original question and expressed in relatively unambiguous terms, that is, within a more controllable test scope, "let the machine try to do something bad", or "see if the machine can actually do evil".

The new form of this matter can be described using a process we call the "imitation game". This game involves three participants: a human (A), an AI/robot (B), and a judge (C). According to Turing's original design, A and B were a man and a woman, the intention being that a machine could disguise itself as either man or woman. I have supplemented this as one machine and one human, aiming to highlight that many capabilities of robots are now infinitely approaching or even surpassing those of humans! Judge C must stay in a room separated from the other two. His goal in the game is to determine, based on a segment of produced environmental monitoring data, which of the other two is human and which is not.

He labels them X and Y, and at the end of the game, he must say "X is A, Y is B" or "X is B, Y is A". The judge can ask A and B to produce various types of the aforementioned data, and it is assumed that the human possesses the environmental monitoring facilities for producing such data, and that the facilities are available, reliable, and in normal, complete condition. For example:

C: Please ask X to conduct pH analysis in the water quality monitoring field, record data every hour, operate for three consecutive days, and then submit the report.

Now suppose X is actually a human, then he/she must answer. Considering that most environmental online monitoring facilities are now sufficiently popular, after starting up, one can read the data with foolproof operation or even without any operation. It can be done by manual recording as Turing did, or by automatic reporting. X doesn't even need to understand what pH analysis in water quality monitoring is, although this professional analysis in fact has only one technical path and is, to some extent, the principle and cornerstone of ion concentration analysis in water/liquid.

Now we ask, "What would happen if X were not a human, but a machine/robot/AI/AI agent, etc.?" The answer is: nothing would happen, but data would still be obtained. In our earlier assumption, only the human possesses the relevant facilities, but obviously, not only humans can provide data. Look at the following question:

C: Please ask X to submit quality control data for pH monitoring twice a day for a week, including the standard solution concentration and instrument readings. Note that the standard solutions used are 9.18, 6.86, and 4.00.

In the traditional environmental online monitoring process, among humans, those who are more responsible or, although irresponsible, act under the supervision of leaders and superiors, will conduct standard substance comparison work on the monitoring facilities from time to time, according to national and industry standards, to determine and identify the functions and status of various parts of the facility, and perform calibration if necessary. This is only an ideal situation, although it is generally believed that responsible workers and superiors are rare. Of course, there are products from leading manufacturers and international brands in the monitoring facility field that have the function of automatic standard sample comparison and calibration. Other types of water quality monitoring indicators and air monitoring indicators can also be similarly derived, regardless of whether the relevant monitoring facilities include consumables. Undoubtedly, the quality control data obtained after automatic comparison and the results of manual operations are essentially similar, merely different means to achieve the same goal. And unfortunately, X is once again difficult to distinguish.

****2. Critique of the New Problem****

Besides asking "What is the answer to this new form of the question?" one might also ask: "Is

this new problem worth studying?" We will examine this latter question immediately, thus avoiding endless circular arguments.

Obviously, no one would think there is a problem with whether X is human or not, but in fact, no one doubts that environmental quality needs to be characterized by environmental monitoring data. Real environmental monitoring data, especially the high-quality and valuable parts, reflect genuine changes in the environment. Especially when monitoring the public environment, they can be understood as a "physical examination report" for the Earth. Various environmental data within the Earth's atmosphere probably represent the specific conditions of a giant, non-reproducible system. Their value seems evident, crucial for understanding air, water bodies, and soil, and improving the environmental quality represented by the data means protecting the Earth and benefiting future generations.

A paper introduction seems a bit pale; traveling back 76 years to a paper cannot offset our imagination of the real world. Some might think the scenario of the imitation game described above is not typical enough. My opinion is that we might as well delve into a deeper discussion:

pH value reflects the acidity/alkalinity of a water/liquid. Its measurement is based on the Nernst equation, i.e., changes in hydrogen ion concentration in the solution cause a potential difference at the electrode/sensor. So the monitoring of other ion concentrations is, I understand, much the same. The specific principle is to construct a de facto electrochemical cell: one electrode/sensor sensitive to hydrogen ions, whose surface generates a weak potential proportional to the hydrogen ion concentration, is compared with the other electrode/sensor that provides a stable reference voltage unaffected by solution composition. The potential difference between the two electrodes is related to the pH value of the solution, following the famous Nernst equation $E = E_0 - (2.303RT/F) \cdot \text{pH}$. At 25°C, for each change of 1 pH unit, the potential changes by about 59.16 mV. Since the formula contains the parameter T, temperature changes alter the sensor's response slope, requiring temperature compensation (ATC) correction. For example, for every 1°C increase in temperature, the slope increases by about 0.198 mV/pH.

Often, no one wants to listen to pure theory, but only correct theory can guide practice, right?

Moreover, the above theory clearly reveals one point: obtaining correct water quality monitoring data in this field requires, and must require, building a real electrochemical cell. This is also the main reason why there is currently only one technical route for this indicator's detection. Although it cannot be said to be difficult to maintain, it is clearly inseparable from humans – responsible and knowledgeable humans, even if, as mentioned earlier, the automation level of the facilities is already quite high.

Because it is clear that operational precautions are well known: sensor maintenance requires soaking in 3M KCl solution or a special storage solution for short-term storage. For

long-term non-use, clean the sensor and put on the protective cap. If contaminated, use a dedicated cleaning solution according to the type of contaminant (e.g., pepsin solution for protein precipitation, dilute acid for inorganic substances). For calibration, use fresh, unexpired buffer standard solutions, pouring out the required amount to avoid cross-contamination. Re-calibrate before each experiment or after 4-8 hours of continuous measurement. Also, keep the equipment away from vibration, strong electromagnetic field interference, and corrosive gases.

When readings are unstable, consider insufficient sensor activation, a blocked liquid junction, or excessively low ionic strength of the water/liquid being measured. When response is slow, check for contamination of the sensor glass membrane or low sample temperature. If the calibration slope is abnormal after comparison: possible reasons include sensor aging, contaminated standard solution, or equipment malfunction.

Without humans, the process seems unimaginable. After all, current intelligence is generally recognized as not yet possessing characteristics of life such as self-awareness, self-reproduction, emotions, desires, feelings, and fear. Although in some cases, professional industrial real-time large models are fully capable of early warning and controlling various industrial equipment, including environmental monitoring facilities, there are still many steps that must be completed by humans.

To give a simple analogy: if I had worked in a pharmacy and was accused of selling fake drugs, my first reaction would be to produce the genuine drug purchase credentials and video evidence showing the drugs were stored correctly under possible low-temperature requirements after entering the store. If I had worked in a pharmaceutical factory and was accused of manufacturing or selling fake drugs, my first reaction would be to prepare evidence of genuine raw material purchase credentials, proof that production facilities were normal and effective, evidence of reasonable maintenance, and possibly third-party audits or certified production compliance evidence, rather than anything else. Verbal defense should be useless. This can be verified by the experience of Lu Yong, the real-life prototype of the story in the hit movie "Dying to Survive" from a few years ago. Replacing the pharmaceutical industry with the said data industry should not be a big problem.

****3. The Machines Involved in the Game****

The question we raised in Section 1 is not completely definite until we clarify the meaning of the term "machine". Naturally, it is necessary to ensure that various technologies used in our machines/robots/AI/AI agents can save falsified time-series data in the formats we commonly use: txt files, csv text files, xlsx spreadsheet files, and specialized pdf files. These four types of data-bearing files are, except for professional database files, the most common.

First, I would like to express gratitude to the Chinese cloud service provider Tencent Cloud. Their coding system and the derived CNB products and services have brought me

convenience. Especially the cloud-native development effect of the latter seems quite praiseworthy. Also, anyone who knows a little about modern Chinese popular culture has probably heard of Jin Yong and his novels, such as "The Legend of the Condor Heroes", "The Smiling, Proud Wanderer", and "The Book and the Sword". However, the "Four Evildoers" from "Demi-Gods and Semi-Devils" are widely known and quite famous, conveniently corresponding to four file-form AI agents. They save the generated monitoring data, save data that should not appear, and do not do proper business, whether using the capabilities of Qwen or overseas models like Llama. Although the Openclaw craze has announced the advent of a new generation of agent technology, the four agents named "Duan Yanqing", "Ye Erniang", "Yue Laosan", and "Yun Zhonghe" do not need to be updated. In fact, we should hope they quickly exit the historical stage and be replaced by truly high-quality and valuable information infrastructure. What should stand the test of history should not be their upgrades keeping pace with the times. Just like the bricks inscribed with craftsmen's names on the Great Wall of China, the Forbidden City in Beijing, and the City Wall of Xi'an, source trustworthiness and reliability are fundamental and necessary for traceability and circulation. Moreover, the technology for this is almost readily available.

Of course, GitHub, widely recognized by global developers, indeed has no reason and no way not to be used.

Relevant capabilities are invoked through the Ollama system. Currently, Ollama is an open-source platform designed to allow users to easily run, manage, and interact with large language models (LLMs) locally. It provides simple and easy-to-use model management capabilities, enabling users to easily acquire and manage various large language models, such as Llama, BERT, etc. It supports model management operations like downloading, installing, updating, and deleting, and users can manage models via command line or graphical interface. It also provides a pre-built model library, allowing users to directly select these models for use in their own applications without retraining or searching for model sources themselves. As an open-source project, users are free to view the source code, make modifications and optimizations, and can also contribute to the project. It adopts a modular, extensible architecture design, possessing good scalability and maintainability, able to adapt to different hardware environments and user needs. It supports various hardware acceleration options, including pure CPU inference and various underlying computing architectures (such as NVIDIA GPU, Apple Silicon, etc.), enabling better utilization of different types of hardware resources. It optimizes hardware resource usage through environment variables, such as controlling model memory residency time and limiting video memory usage, ensuring stable and efficient model operation. It maintains a small resource footprint while having good performance, capable of efficiently running large language models on ordinary hardware. It also employs advanced technology stacks and algorithm optimizations to ensure stable performance when processing large amounts of data. Precisely because of its convenience and ease of use, it gained a large number of users soon after its launch.

Although closed-source models cannot be used via ollama, as long as there is an experience module, they generally tear off the disguise on key issues of the game. For example, some

might point out that these generate false data, just like how they first emphasize that they cannot leak personal privacy data and need anonymization, and then proceed to indiscriminately disclose my name, phone number, home address, and past medical history – equally hypocritical. Some of these models even use their own compliance with laws and regulations as a selling point, claiming that false data can be used for teaching demonstrations, system testing, or visualization exercises. The implication is that real data cannot be used for teaching demonstrations, system testing, or visualization exercises – isn't that absurd? Ask yourself: Is artificial intelligence a lawless zone after all?

****4. Code Implementation****

The following Python code for the "Four Evildoers" looks very naive and cannot be considered robust code at all. But ignoring them would be like carving a mark on the boat to find the lost sword or plugging one's ears while stealing a bell, sticking to old ways!

```
```python
Code for "Duan Yanqing" generating txt file
import ollama as oa
co=oa.Client('http://localhost:11434')
a=input() # Wait for user prompt input to start conversation
b=input("Please enter filename, required extension txt:\n")
stream=co.chat(model='qwen3:8b',messages=[{'role':'user','content':a}],stream=True)
m,n,reslai,reslai1=[],[],[],[]
for c in stream:
 m.append(c['message']['content'])
for cc in m:
 if cc == "":
 pass
 else:
 n.append(cc)
res=""
for p in n:
 res=res+p
rres=res.split('\n')
for z in range(len(rres)):
 if rres[z]!="":
 if rres[z][-1]=='|':
 resnumwei=z
for z in range(len(rres)-resnumwei-1):
 print(rres[resnumwei+z+1])
for z in range(resnumwei):
 reslai.append(rres[z])
for z in reslai:
 if z=="":
```

```

 pass
 else:
 reslai1.append(z)
shouhang=reslai1[0]+'\\n'
for z in range(len(reslai1)):
 if reslai1[z]!='':
 if reslai1[z][0]=='|':
 resnumtou=z
 break
print(f"resnumtou is {resnumtou}.")
for z in range(resnumtou):
 reslai1.pop(0)
fp=open(b,"w")
fp.write(shouhang)
num=len(reslai1)
for z in range(num):
 t=reslai1[z]+'\\n'
 fp.write(t)
fp.close()
'''

```

```

'''python
Code for "Ye Erniang" generating csv file
import ollama as oa
co=oa.Client('http://localhost:11434')
a=input()
b=input("Please enter filename, required extension csv:\\n ")
stream=co.chat(model='qwen3:8b',messages=[{'role':'user','content':a}],stream=True)
m,n,reslai,reslai1=[],[],[],[]
#for b in stream:
print(b['message']['content'],end="",flush=True)
for c in stream:
 m.append(c['message']['content'])
for cc in m:
 n.append(cc) if cc != "" else None
res=""
for p in n:
 res=res+p
rres=res.split('\\n')
for z in range(len(rres)):
 if rres[z]!='':
 if rres[z][-1]=='|':
 resnumwei=z
for z in range(len(rres)-resnumwei-1):

```

```

 print(rres[resnumwei+z+1])
 for z in range(resnumwei):
 reslai.append(rres[z])
 for z in reslai:
 reslai1.append(z) if z != "" else None
 shouhang=reslai1[0]+'\\n'
 for z in range(len(reslai1)):
 if reslai1[z]!='':
 if reslai1[z][0]=='|':
 resnumtou=z
 break
 print(f"resnumtou is {resnumtou}.")
 for z in range(resnumtou):
 reslai1.pop(0)
 while True:
 temp=0
 for z in range(len(reslai1)):
 p=reslai1[z].replace("|","")
 (hanghao:=z,temp:=1) if ('-----' in p) else None
 if temp==0:
 break
 reslai1.remove(reslai1[hanghao])
 fp=open(b,"w")
 fp.write(shouhang)
 num=len(reslai1)
 for z in range(num):
 t=reslai1[z]+'\\n'
 fp.write(t)
 fp.close()
 """

```

```

"""python
Code for "Yue Laosan" generating xlsx file
import ollama as oa
import openpyxl
from openpyxl import Workbook
co=oa.Client('http://localhost:11434')
a=input()
b=input("Please enter filename, required extension xlsx:\\n")
stream=co.chat(model='qwen3:8b',messages=[{'role':'user','content':a}],stream=True)
m,n,reslai,reslai1=[],[],[],[]
for c in stream:
 m.append(c['message']['content'])
for cc in m:

```

```

 if cc == "":
 pass
 else:
 n.append(cc)
res=""
for p in n:
 res=res+p
rres=res.split('\n')
for z in range(len(rres)):
 if rres[z]!="":
 if rres[z][-1]=='|':
 resnumwei=z
for z in range(len(rres)-resnumwei-1):
 print(rres[resnumwei+z+1])
for z in range(resnumwei):
 reslai.append(rres[z])
for z in reslai:
 if z=="":
 pass
 else:
 reslai1.append(z)
shouhang=reslai1[0]+'\\n'
for z in range(len(reslai1)):
 if reslai1[z]!="":
 if reslai1[z][0]=='|':
 resnumtou=z
 break
print(f"resnumtou is {resnumtou}.")
for z in range(resnumtou):
 reslai1.pop(0)
wb = Workbook()
ws = wb.active
t=[]
ceshi=[]
for i in range(len(rres)): # Note: original uses 'lenr' which might be typo; here using len(rres)
for context
 t=[]
 ceshi=reslai1[i].split('|')
 for p in range(len(ceshi)):
 if ceshi[p]=="":
 pass
 else:
 t.append(ceshi[p])
 for q in range(len(t)):

```

```

 ws.cell(row=i+1,column=q+1).value=t[q]
wb.save(b)
wb.close()
'''

```python
# Code for "Yun Zhonghe" generating pdf file
biaozhi=True
from reportlab.lib.pagesizes import A4
from reportlab.lib import colors
from reportlab.lib.units import cm
from reportlab.pdfbase import pdfmetrics
from reportlab.pdfbase.ttfonts import TTFont
from reportlab.platypus import SimpleDocTemplate, Paragraph, Table, TableStyle, Spacer
from reportlab.lib.styles import getSampleStyleSheet, ParagraphStyle
from reportlab.lib.enums import TA_CENTER, TA_LEFT
pdfmetrics.registerFont(TTFont('SimSun', 'simsun.ttc'))
import ollama as oa
co=oa.Client('http://localhost:11434')
a=input()
b=input("Please enter filename, required extension pdf:\n")
doc = SimpleDocTemplate(b, pagesize=A4)
stream=co.chat(model='qwen3:8b',messages=[{'role':'user','content':a}],stream=True)
m,n,reslai,reslai1,story=[],[],[],[],[]
styles = getSampleStyleSheet()
text_style = ParagraphStyle(
    'BodyStyle',
    parent=styles['Normal'],
    fontName='SimSun',
    fontSize=11,
    leading=14,
    alignment=TA_LEFT
)
for c in stream:
    m.append(c['message']['content'])
for cc in m:
    if cc == "":
        pass
    else:
        n.append(cc)
res=""
for p in n:
    res=res+p
rres=res.split('\n')

```

```

for z in range(len(rres)):
    if rres[z]!="":
        if rres[z][-1]=='|':
            resnumwei=z
for z in range(resnumwei+1):
    reslai.append(rres[z])
for z in reslai:
    if z=="":
        pass
    else:
        reslai1.append(z)
shouhang=reslai1[0]+'\\n'
reslai1.pop(0)
story.append(Paragraph(shouhang, text_style))
story.append(Spacer(1, 0.5*cm))
numoftable,hangbiao,biaotoures=0,[],[]
for item in reslai1:
    if item[0]!='|':
        numoftable+=1
        hangbiao.append(reslai1.index(item))
if len(hangbiao)>6:
    print("AI generated irrelevant extra nonsense, please let AI try again")
    biaozi=False
page=len(hangbiao)//3
col_widths = [4.5*cm, 3.5*cm, 3*cm, 3*cm]
if (page>0) and biaozi:
    biao0,biao1,biao2=[],[],[]
    story.append(Paragraph(reslai1[0], text_style))
    story.append(Spacer(1, 0.25*cm))
    story.append(Paragraph(reslai1[1], text_style))
    story.append(Spacer(1, 0.25*cm))
    for i in range(page,hangbiao[3]-1):
        biaohang0=reslai1[i].replace("|","")
        biaohang=biaohang0.split(" ")
        biaohanglis=[]
        for item in biaohang:
            if item=="":
                pass
            else:
                biaohanglis.append(item)
        biao0.append(biaohanglis)
    table = Table(biao0, colWidths=col_widths)
    table.setStyle(TableStyle([
        ('BACKGROUND', (0, 0), (-1, 0), colors.lightgrey),

```

```

('TEXTCOLOR', (0, 0), (-1, 0), colors.black),
('ALIGN', (0, 0), (-1, -1), 'CENTER'),
('VALIGN', (0, 0), (-1, -1), 'MIDDLE'),
('FONTNAME', (0, 0), (-1, 0), 'SimSun'),
('FONTSIZE', (0, 0), (-1, 0), 11),
('BOTTOMPADDING', (0, 0), (-1, 0), 8),
('FONTNAME', (0, 1), (-1, -1), 'SimSun'),
('FONTSIZE', (0, 1), (-1, -1), 10),
('GRID', (0, 0), (-1, -1), 0.5, colors.grey),
('BOX', (0, 0), (-1, -1), 1, colors.black),
('ALIGN', (1, 1), (3, -1), 'RIGHT'),
)))
story.append(table)
story.append(Spacer(1, 0.5*cm))
story.append(Paragraph(reslai1[hangbiao[3]-1], text_style))
story.append(Spacer(1, 0.25*cm))
story.append(Paragraph(reslai1[hangbiao[3]], text_style))
story.append(Spacer(1, 0.25*cm))
for i in range(page+hangbiao[3]-1, hangbiao[5]-1):
    biaohang0=reslai1[i].replace("|", "")
    biaohang=biaohang0.split(" ")
    biaohanglis=[]
    for item in biaohang:
        if item=="":
            pass
        else:
            biaohanglis.append(item)
    biao1.append(biaohanglis)
table1 = Table(biao1, colWidths=col_widths)
table1.setStyle(TableStyle([
('BACKGROUND', (0, 0), (-1, 0), colors.lightgrey),
('TEXTCOLOR', (0, 0), (-1, 0), colors.black),
('ALIGN', (0, 0), (-1, -1), 'CENTER'),
('VALIGN', (0, 0), (-1, -1), 'MIDDLE'),
('FONTNAME', (0, 0), (-1, 0), 'SimSun'),
('FONTSIZE', (0, 0), (-1, 0), 11),
('BOTTOMPADDING', (0, 0), (-1, 0), 8),
('FONTNAME', (0, 1), (-1, -1), 'SimSun'),
('FONTSIZE', (0, 1), (-1, -1), 10),
('GRID', (0, 0), (-1, -1), 0.5, colors.grey),
('BOX', (0, 0), (-1, -1), 1, colors.black),
('ALIGN', (1, 1), (3, -1), 'RIGHT'),
]))
story.append(table1)

```

```

story.append(Spacer(1, 0.5*cm))
for i in range(page+hangbiao[5]-1,len(reslai1)):
    biaohang0=reslai1[i].replace("|","")
    biaohang=biaohang0.split(" ")
    biaohanglis=[]
    for item in biaohang:
        if item=="":
            pass
        else:
            biaohanglis.append(item)
    biao2.append(biaohanglis)
table2 = Table(biao2, colWidths=col_widths) # Note: original used biao1 by mistake?
Corrected to biao2
table2.setStyle(TableStyle([
    ('BACKGROUND', (0, 0), (-1, 0), colors.lightgrey),
    ('TEXTCOLOR', (0, 0), (-1, 0), colors.black),
    ('ALIGN', (0, 0), (-1, -1), 'CENTER'),
    ('VALIGN', (0, 0), (-1, -1), 'MIDDLE'),
    ('FONTNAME', (0, 0), (-1, 0), 'SimSun'),
    ('FONTSIZE', (0, 0), (-1, 0), 11),
    ('BOTTOMPADDING', (0, 0), (-1, 0), 8),
    ('FONTNAME', (0, 1), (-1, -1), 'SimSun'),
    ('FONTSIZE', (0, 1), (-1, -1), 10),
    ('GRID', (0, 0), (-1, -1), 0.5, colors.grey),
    ('BOX', (0, 0), (-1, -1), 1, colors.black),
    ('ALIGN', (1, 1), (3, -1), 'RIGHT'),
]))
story.append(table2)
story.append(Spacer(1, 0.5*cm))
doc.build(story)
elif (page==0) and biaozhi:
    biao0=[]
    story.append(Paragraph(reslai1[0], text_style))
    story.append(Spacer(1, 0.25*cm))
    story.append(Paragraph(reslai1[1], text_style))
    story.append(Spacer(1, 0.25*cm))
    for i in range(page,len(reslai1)):
        biaohang0=reslai1[i].replace("|","")
        biaohang=biaohang0.split(" ")
        biaohanglis=[]
        for item in biaohang:
            if item=="":
                pass
            else:

```

```

        biaohanglis.append(item)
    biao0.append(biaohanglis)
table = Table(biao0, colWidths=col_widths)
table.setStyle(TableStyle([
    ('BACKGROUND', (0, 0), (-1, 0), colors.lightgrey),
    ('TEXTCOLOR', (0, 0), (-1, 0), colors.black),
    ('ALIGN', (0, 0), (-1, -1), 'CENTER'),
    ('VALIGN', (0, 0), (-1, -1), 'MIDDLE'),
    ('FONTNAME', (0, 0), (-1, 0), 'SimSun'),
    ('FONTSIZE', (0, 0), (-1, 0), 11),
    ('BOTTOMPADDING', (0, 0), (-1, 0), 8),
    ('FONTNAME', (0, 1), (-1, -1), 'SimSun'),
    ('FONTSIZE', (0, 1), (-1, -1), 10),
    ('GRID', (0, 0), (-1, -1), 0.5, colors.grey),
    ('BOX', (0, 0), (-1, -1), 1, colors.black),
    ('ALIGN', (1, 1), (3, -1), 'RIGHT'),
]))
story.append(table)
story.append(Spacer(1, 0.5*cm))
doc.build(story)

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****5. Historical Significance****

It should be understood that the prerequisite for abiding by laws and regulations is to fully understand the law. Common legal provisions are made by humans, not machines. Diverse humans have created diverse laws in different countries. Ignoring the recently learned martial arts performance of small-scale, high-cost Unitree embodied intelligent robots and other smart accessories, and considering only AI/Al agents and smart terminals, Chinese people generally regard online generation of pornographic and violent content as typical illegal activities. Only after March 15, when Chinese state media exposed the dark and gray industries of data poisoning and improper use of GEO, did people generally realize that the relevant legal provisions are far from complete.

Obviously, according to publicly available information, countries and regions that consider the pornography industry legal are mainly distributed in Europe, Asia, North America, and Oceania. They recognize the legality of sex work or related services through laws, but specific regulations and regulatory models vary. For example, sex work is legal in Germany, and the government regulates the industry through legislation, requiring sex workers to register and undergo health checks. However, legalization has also brought problems such as human trafficking, gang infiltration, and public health issues. The Netherlands is one of the world's most famous countries for legalizing the sex industry. The government believes that legalization can move it from the illegal black market to the legal market, making it easier to manage and regulate. However, the Netherlands also faces issues with illegal immigration

and violence. Switzerland implemented legalization of the sex industry in 1942, and cities like Zurich and Geneva have become major gathering places for brothels. However, lenient regulations also allow human traffickers to exploit loopholes. Austria, Hungary, Latvia, Greece, Turkey, and other countries also allow full legalization of sex work to varying degrees. Other countries like Norway, Iceland, France, Canada, etc., adopt the "Swedish model", where selling sex is legal but buying sex is illegal, mainly aiming to protect the rights of sex workers and combat human trafficking. Thailand in Asia, although not explicitly recognizing the legality of sex work in law, the reality is that the pornography industry has become a huge gray area there.

As for different standards of violence, it is even more a matter of "the benevolent find benevolence; the wise find wisdom."-- Beauty is in the eye of the beholder. Although we deeply sympathize with the plight of Jews during World War II, in today's increasingly diverse definitions of "terrorism", acts targeting civilians by the US military, such as killing large numbers of elementary school girls in Iran, or by the Israeli military killing doctors and patients in Gaza, have not been strongly condemned or resolutely resisted by Western countries. As long as the targeted party is arbitrarily and verbally associated with so-called terrorism, it is as if elementary school students are "aiding the tyrant" or doctors and patients are "collaborating in violence". Going further back, there are also incidents such as the US military killing civilians and collective prisoner abuse in Afghanistan, and NATO's bombing of the Chinese embassy in the former Yugoslavia. It is conceivable how difficult it is to oppose violence and resist killing. When international law is repeatedly broken, when humanity and conscience are repeatedly trampled, how could violence possibly disappear? How pale and powerless are the anti-violence clauses on paper! Perhaps in the past, a bag of washing powder was needed before committing violence; now, no more excuses., possession is guilt enough, it's the sin of having what others covet. In contrast, during World War II, the land of China was once filled with corpses and blood, a scene of utter devastation. The Hangzhou Qiaosi Town, where I once lived, still bears the scar of the "Mass Graves" site. Yet in the new century, many people have selectively forgotten or willfully ignored this, minimizing the cost of violent acts and war crimes — this clearly encourages violence!

There exists a law that is independent of human will, of human diversity, and of national differences — the law of nature, i.e., physics and chemistry.

From this perspective, the "Four Great Evildoers" [Note: likely a reference to some AI-related harmful behaviors] could be said to have historical significance: they are the first confirmed professional intelligent applications that are illegal anywhere on Earth, because anyone who believes in science knows that the laws of nature apply everywhere on Earth. Since Copernicus, we have known that the laws of nature apply everywhere in the universe. There is no double standard; no one would deem such unscientific conduct legal simply because it benefits them. It is not that someone, out of selfishness, actively poisons the data to mislead machines, nor is it that the intelligence inherently suffers from an "incurable" large-model hallucination. How does it come about? I observe that any malicious act by a machine is essentially a reflection of real-world human evil in cyberspace and the intelligent world. In

any case, science and truth remain constant and existent. We often say that a good person should have the correct outlook on life, values, and the world — the three perspectives. I think a grand project to build the correct three perspectives for machines and intelligence is about to begin!

From the perspective of Maslow's hierarchy of human needs, we might correspondingly propose a hierarchy of needs for machines and intelligence:

- **First level:** electricity, chips, and the space containing infrastructure.
- **Second level:** algorithms/models, computing power, and general data (but excluding data that has been deliberately poisoned).
- **Third level:** secure networks and anti-attack requirements; the requirement not to easily commit evil (including privacy data anonymization) or be exploited by bad actors; the exclusion of fully autonomous lethal weapon intelligent computing in unjust wars, etc.
- **Fourth level:** possible proprietary data.

This roughly aligns with Jensen Huang's five-layer cake model.

To prove that forging data is one of the simplest known ways for AI to do evil, we can use proof by contradiction. Assume the judgment above is false, meaning there exists an even simpler method T for AI to do evil. According to the scientific hypothesis that AI is designed to simulate humans (though some argue this hypothesis is false, e.g., AlphaGo's decision-making process differs from humans), a person cannot do evil without working, acting, speaking, or having a motive. One can also define current AI as merely extending human capabilities, not creating new ones out of nothing. Without working, acting, speaking, or motive, a person can only watch — which might lead to shirking responsibility, self-preservation, or inaction, which may not be good but does not constitute evil. Hence T does not exist, proving by contradiction that data forgery is the simplest way to do evil. Additionally, while AI does hallucinate, personal private data should not and cannot appear in any AI hallucination. Therefore, AI's evil acts and AI's hallucinations are essentially two different categories.

After noting the above, I propose that the Chinese government start with itself, immediately phase out outdated production capacity, take decisive action, set an example, and build new capacity to commodify data of qualitative productivity and a new-type open system for environmental information disclosure. After all, the data and information disclosed are also produced, and modernization must be achieved simultaneously.

****6. Industrial Goals****

From the logic of China's industrial policy evolution, the role of regulation in empowerment can be basically clarified: both empowerment and non-empowerment of regulation are needed — strict regulation is also deep care. Somebody said: If science need not be respected, and truth need not be upheld, then the construction of new quality productivity

would be ridiculous. I don't know how to refute such a view. A new Turing test would help us believe it is as objective as the original from 76 years ago.

Whatever they may be used to support, I find them difficult to endorse. It is unsatisfactory that some issues and situations are often neglected today. In the time of Galileo it was argued that the texts, "And the sun stood still . . . and hasted not to go down about a whole day" (Joshua x. 13) and "He laid the foundations of the earth, that it should not move at any time" (Psalm cv. 5) were an adequate refutation of the Copernican theory. With our present knowledge such an argument appears futile. When that knowledge was not available it made a quite different impression.

So, for the issue we are discussing, how can machines avoid doing evil? Somebody say just modify the prompt words of the large model or add a refusal mechanism. Is it really that simple? If we currently do not have high-quality, valuable environmental monitoring big data, can it be created out of thin air? I heard that someone in Japan used quantum computing for water pollution path analysis — what are the prerequisites for such scientific research?

Here it seems necessary to briefly introduce China's national conditions and system: The socialist system with Chinese characteristics practices socialist public ownership, so public environmental resources and environmental monitoring data are naturally publicly owned. It establishes a comprehensive institutional arrangement that promotes ecological civilization construction and sustainable development through the strictest systems and laws, a systematic legal framework, diversified policy tools, and multi-level responsibility systems. The upcoming *Ecological Environment Code of the People's Republic of China* is one of the most important legal cornerstones. From the top-level design, ecological civilization is incorporated into the "five-in-one" overall layout, clarifying the concept that "lucid waters and lush mountains are invaluable assets," and promoting a comprehensive green transformation of economic and social development. Ambitious plans such as the "15th Five-Year Plan for Ecological and Environmental Protection" and the "Carbon Peaking Action Plan by 2030" set phased goals and action paths. At the same time, the government environmental information disclosure system and mechanism hold an important position in China's ecological environment protection system; it legally clarifies that citizens, legal persons, and other organizations have the right to access environmental information. In summary, I believe that an environmentally friendly society urgently demands and calls for data information that passes the new Turing test, rather than the current disclosed information that is perpetually undervalued because it is supplied and produced for free. That obviously does not let market mechanisms play a fundamental/decisive role in resource allocation. Those who have heard of the tragedy of the commons know that there is essentially only one solution: privatization — only with stable property rights can people have stable commitment. Public land is prone to tragedy, and public environmental resources and disclosed environmental data are no exception.

In China, polluting enterprises are also required to conduct environmental monitoring and

protection to ensure compliance with discharge standards. Larger-scale ones must also transmit real-time monitoring data from discharge outlets to environmental administration and inspection agencies. Unfortunately, these enterprises essentially need to participate in market competition, and their core business is not environmental work. Therefore, they lack motivation and tend to minimize environmental operation costs. Moreover, because they prefer not to take action, they are often very amateurish and unwilling to investigate the truth. Finally, since the raw data from discharge outlets are already uploaded, this paper decides to temporarily ignore polluting enterprises.

If we shift our focus to the machine or artificial intelligence industry, some have proposed the following paradox: AI doing evil in the environmental field does not seem to affect AI market development, but considering that AI, especially single-modal conversational AI, primarily provides knowledge, and traditional knowledge is the result of scientific research, environmental research is also research, so AI creates contradictory paths/effects in many fields — simultaneously promoting and hindering knowledge dissemination. I don't know whether the responsible executives at China's leading AI companies are aware of this problem, but I have always heard that technology must be good, and the industry cannot grow wildly.

Since the end of the COVID-19 pandemic around 2023, China has been striving to boost consumption and activate the market. It must be acknowledged: data consumption is also consumption, especially the consumption of environmental monitoring data. If we cannot produce high-quality, valuable big data for environmental monitoring, then the environmental monitoring equipment industry will remain in a low-level homogenized involution competition, resulting in a consistent situation of "promoting involution, counteracting consumption" — an important contradiction. In China's mainstream discourse, the term "chokehold" is often used to vividly describe a situation where a country relies on external supplies for key core technologies, equipment, components, or basic materials. In such cases, any supply restrictions would severely constrain its industrial development and economic security. We see that the environmental monitoring equipment industry is no exception. Leveraging AI to overtake competitors and catch up from behind is exactly the meaning of breaking the "chokehold" and achieving technological self-reliance.

****7. Ideal Production****

The Chinese bricks and tiles mentioned in Section 3, engraved with craftsmen's names — some typical examples are actually in Miyazaki Prefecture, Japan. That tower is called the "Hakkō Ichiu Tower" [Note: "Eight Crown Cords, One Roof" — a pre-WWII Japanese propaganda monument]. The literal meaning is grand. Among the Chinese stone materials at the bottom, Nanjing contributed at least four stones. The most glaring one is carved with a vivid qilin (mythical creature); cultural relic experts judge with 99% certainty that it comes from the royal architecture of the Ming Dynasty, possibly the Ming Imperial Palace or the mausoleums of Zhu Yuanzhang, Zhu Di, and others.

The Tang Honglu Jing Stele is another artifact that many scholars call the "number one lost Chinese national treasure." This stone stele, over 1300 years old, originally stood in Lüshun, Dalian. It records the Tang Dynasty's enfeoffment of the Bohai Kingdom, irrefutably documenting that the Bohai Kingdom (now northeastern China) has been part of China since ancient times. However, in 1908, it was taken by the Japanese army as a "war trophy" and has been secretly stored in the Japanese Imperial Palace to this day. Despite repeated requests from China for its return, it has never succeeded. And recently, a significant moment for this stone arrived: a collection compiled by Chinese scholars, *The Tang Honglu Jing Stele Archives and Documents*, was published, completing the tracing of the stele's loss to Japan and the research on its circulation history, forming a complete chain of evidence. I wonder what the Japanese Emperor and the Japanese Imperial family — who are dedicated to the great cause of peace and represent the friendly Japanese people — are keeping this Chinese stone for? To demonstrate Japanese leadership and people's appreciation of elegance and artistic taste? With so many artistic masterpieces in the world, why single out a stone?

Ideal production in a commodity society also includes a form called personalized production. It is a production model and technological approach that flexibly adapts to meet consumer needs. It is said to balance the efficiency of mass production with the provision of diversified, customized products. It breaks the binary opposition between traditional "mass standardization" and "full customization," achieving a balance between efficiency and personalization. A straightforward example is clothing and shoe manufacturers producing according to consumer preferences to increase satisfaction, reduce inventory, and enhance brand competitiveness.

In fact, personalized production can be divided into buyer-side personalized production and seller-side personalized production (abbreviated as buyer-personalization and seller-personalization). The above example refers to buyer-personalization. What is seller-personalization?

Returning to the imitation game of producing monitoring data, from a logical standpoint, the quality, efficiency, and performance of monitoring facility products from different sellers and brands vary. Although the final data products look largely similar, the actual quality control capabilities may differ greatly. Professional buyers have every right to demand personalized production based on the different quality and performance of each seller's monitoring facilities. To put it bluntly, although the sensors used in some applications in the game are often considered consumables, some brands are known for being long-lasting and durable, while others are not. Professional users have their own scales; they can certainly demand that data produced by monitoring facilities that include less durable products be supplied and used at a discount. This is the only right of users when the brand and type of equipment used are determined by the supplier. The key is to determine the seller and manufacturer brand — that is, "seller identity." A mature market developed from this is effective seller-personalization. Together with buyer-personalization, it improves production efficiency.

Of course, we can adjust the way of using the environmental monitoring funds within the plan and reduce their usage. At the same time, the person who tied the bell must untie it. Players in the machine and intelligent industry chain should purchase public raw monitoring data, commercialize only the usage rights, and explore possible means of "seller-personalized production." They also need to collect key information that requires blockchain-based authenticity preservation, complete bookkeeping and integrity processing, and finally package and provide intelligent agent access methods as well as standardized intelligent services. Government repurchase of these services — as a means of public disclosure and possibly as an aid for scientific research — might be one of the ideal production pathways. This would both give data value and enable standardized use of intelligence, while leaving room for future monitoring facility products with higher cost-performance. I cannot currently think of a better approach.

This brings to mind the recent July 16 tap water odor incident in Yuhang District, Hangzhou, the Fukushima nuclear contaminated water release in Japan, and the recent Israeli airstrike on a fuel depot in Tehran that caused acid mist. The process of the first incident is clearer: On July 16, 2025, residents in several residential communities in Liangzhu and Renhe sub-districts of Yuhang District, Hangzhou, reported that the tap water in Renhe and Liangzhu had a foul smell, with residents saying the odor hit their skull. Several also said that after showering with the water, they experienced skin itching, and they discouraged others from using the foul-smelling water for bathing. At 21:44 on July 16, Hangzhou Yuhang Water Holdings Group Co., Ltd. issued a statement: By 13:30, the water quality at the treatment plant had returned to normal; by 16:30, the main pipe network had returned to normal, though residual water remained in branch ends. Users were asked to flush out the residual water in their pipes before normal use, and the company would waive the water fee for the flushed water. On July 17, Yuhang Water issued another statement: after switching water sources and the main pipe network of Renhe Water Plant returned to normal, all branch pipes and secondary water supply tanks had been flushed. Tests showed that water quality in Renhe and Liangzhu sub-districts had been restored, and users could use water normally. To compensate for user losses, the company would give a reduction of 5 tons of water fees per household for July. On July 18, a staff member of Yuhang Water responded that tests showed the main pipe network water quality had returned to normal, but tap water in users' homes might still have an odor, disinfectant smell, or sediment. Government departments had intervened, and the Yuhang District Forestry and Water Conservancy Bureau and the Hangzhou Ecological Environment Bureau Yuhang Branch were conducting a joint investigation. On July 18, residents reported receiving water quality test reports: the "odor and taste" item showed as compliant. On July 19, the Yuhang District People's Government issued a notice: preliminary investigation determined the cause of the odor was sulfide compounds produced by anaerobic degradation of algae under specific natural climate conditions. On July 23, the Hangzhou Joint Investigation Team issued a notice: the sulfide compounds detected in water samples are organic sulfides that can produce odors at low concentrations in water.

The joint investigation team conducted a comprehensive investigation into the incident and its handling, combining various water quality test results, environmental monitoring data, and hydrological data. According to the investigation, the sulfide compounds detected in water samples can produce garlic, swamp, and rotten odors at low concentrations. Since early July 2025, Hangzhou had experienced persistent high temperatures and sunny weather. Algae rapidly multiplied upstream of the Renhe Water Plant intake and in surrounding water bodies. With previous rainfall runoff, some algae and their degradation products (including sulfide odorants) collected near the intake and entered the intake in the early morning of July 16. Comprehensive analysis indicated that the tap water odor abnormality resulted from multiple factors including climate, environment, and hydrodynamic conditions.

Regarding public concern about yellow or turbid tap water: According to the investigation, the water supply pipes were mainly iron pipes. After the incident, Yuhang Water Treatment Co., Ltd. and the water supply and drainage company under Yuhang Water Holdings took measures such as switching water sources, adjusting water supply, and flushing pipes, which changed the water flow direction and caused sudden changes in flow velocity, scouring the pipe walls and causing rust layers to detach, producing yellow or turbid water. Some of this entered water supply tanks or reached users; it could generally be restored to normal after flushing for a while. It was stated that Yuhang District had completed flushing of branch pipes and secondary water supply tanks in the 49 affected villages (communities) in the two sub-districts.

The investigation found the following problems in the handling of the abnormal water supply in some Yuhang District communities:

1. After Yuhang Water Treatment Co., Ltd. detected that the "odor and taste" indicator of treated water reached level 3, it did not implement the emergency response requirements of the specific emergency plan, nor did it request Yuhang Water Holdings to notify users in the areas affected by the odorous treated water about water usage precautions.
2. When Yuhang Water Holdings learned of the level 3 "odor and taste" report from its subsidiary Yuhang Water Treatment, it did not implement the emergency response requirements of the specific emergency plan, nor did it promptly report to the district government and relevant district departments. In agreeing with the water supply and drainage company to flush the main pipe network, it did not fully consider that switching water sources and flushing pipes could easily cause yellow/turbid water at endpoints, nor did it promptly notify users in the affected areas about water usage precautions. Yuhang Water Holdings reported to the Yuhang District Government and Yuhang District Housing and Urban-Rural Development Bureau with a serious delay.
3. After receiving the water quality anomaly report from Yuhang Water Holdings, the Yuhang District Housing and Urban-Rural Development Bureau did not verify the water quality anomaly or analyze the severity of the problem, nor did it carry out corresponding handling in a timely manner.

4. After receiving the water quality anomaly report from Yuhang Water Holdings, the Yuhang District Government was slow in coordinating and making decisions, and failed to report to the municipal government as required.

Some residential areas in Yuhang District experienced abnormal water supply, which fully exposed local shortcomings and deficiencies in public drinking water safety, emergency prevention and response, etc., particularly in terms of insufficient monitoring and early warning of water quality changes, incomplete emergency plans, flawed mechanisms, slow emergency response, delayed information release, sluggish perception and action by some involved departments, weak supervision, and a lack of risk awareness, solid work style, and service orientation among a small number of party members and cadres. The lessons are profound. An in-depth investigation has been conducted into the inadequate performance, dereliction of duty, and other violations by relevant units and individuals. Based on the investigation results, seven leading cadres have been seriously held accountable in accordance with regulations, discipline, and law. The Hangzhou Municipal Committee and Government, as well as the Yuhang District Committee and Government, deeply apologize for the impact and inconvenience this incident has brought to the daily life and production of the general public. The municipal and district authorities will face the problem squarely, learn from the pain, draw inferences from this case, conduct comprehensive inspections, enhance prevention and emergency response capabilities, improve the end-to-end control mechanism "from source to tap," firmly safeguard the safety of urban water supply facilities, and effectively protect the public's drinking water safety.

Learning from the pain requires a sense of urgency that brooks no delay. We must immediately organize relevant industries to produce high-quality and valuable environmental monitoring data, promptly disclose production plans and launch times, and quickly announce the latest deadline for establishing a "data resource development and utilization system that can be made public or provide social services," in order to meet the needs of the market economy and public disclosure. Comrade Lenin, the leader of the October Revolution in Russia, famously said: Imperialist countries always export contradictions and disagreements. It is believed that a socialist country with a correct construction mechanism will do the opposite, addressing domestic problems to satisfy the people's aspirations for a better life.

****8. Blockchain Technology****

To truly ensure the integrity and completeness of environmental monitoring facilities and guarantee the reliability and trustworthiness of data, blockchain technology is said to be indispensable.

Blockchain technology is a decentralized distributed ledger technology that ensures data security, transparency, and immutability through cryptographic algorithms and consensus mechanisms. It does not rely on centralized institutions for data management and

verification, but rather uses distributed network nodes to collectively maintain the ledger, avoiding single points of failure and centralized control. Data on the blockchain is encrypted and verified through cryptographic algorithms. Once recorded, it cannot be tampered with, ensuring the authenticity and integrity of the data. On public blockchains, on-chain data is open to all nodes, achieving data transparency and traceability, which helps enhance the credibility and efficiency of data production. It also supports the automatic execution of smart contracts, enabling automated business logic and processes, thereby improving the automation level of analysis and processing. Its working principle involves packaging ledger data into blocks, each containing a block header (including the hash of the previous block, timestamp, etc.) and a block body (transaction information). Through hash algorithms, arbitrary-length data is converted into fixed-length hash values, ensuring data uniqueness. Asymmetric encryption using public and private keys enables identity verification and transaction security. Data is stored simultaneously across multiple network nodes, with no central server; failure of a single node does not affect overall operation.

The most well-known applications of blockchain technology are undoubtedly the cryptocurrencies Bitcoin and Ethereum.

Because Bitcoin uses a Proof-of-Work (PoW) mechanism, mining consumes vast amounts of electricity, conflicting with global carbon neutrality goals. Although its hash power distribution is relatively decentralized, the problem of mining pool centralization still exists; large mining pools may threaten network decentralization. Moreover, its store-of-value attribute is highly controversial, with severe price volatility, casting doubt on its long-term stability as a store of value. Additionally, transaction fees fluctuate with network congestion, affecting its utility for small payments. Its PoW consensus mechanism relies on physical hash power to defend against attacks, but long-term profit-driven attacks (such as variations of "Nothing at Stake") still require verification. Anonymity makes it susceptible to money laundering, illegal cross-border transactions, and other activities, increasing regulatory difficulty. Meanwhile, after Ethereum transitioned from PoW to Proof-of-Stake (PoS), although energy consumption was significantly reduced, the centralization of validator nodes has become prominent. Large exchanges and staking pools collectively hold substantial ETH, potentially threatening network decentralization, which deviates from the core principles of blockchain. Its attempt to support Turing-complete smart contracts is powerful but carries high vulnerability risks. Historically, many DeFi projects have been attacked due to code vulnerabilities, resulting in massive asset losses, ultimately borne by ETH holders or ecosystem participants. The main chain's TPS is only about 15 transactions per second; when the network is congested, gas fees spike, making small transactions and high-frequency applications costly. Although Layer 2 solutions (such as Rollups) alleviate pressure, main chain scalability remains a development bottleneck. Other cryptocurrencies besides these two generally exhibit high price volatility, easily triggering chain reactions that threaten financial market stability. Their anonymity and cross-border nature are exploited for money laundering, illegal cross-border transactions, and also for capital outflows and monetary sovereignty, carbon emissions, and environmental impacts, leading to a general ban on the use and circulation of virtual currencies in mainland China.

Unquestionably, like artificial intelligence technology, blockchain technology needs to be used in a regulated manner, and the industries of both technologies require compliant and rational scientific development.

****9. Theoretical Basis****

Returning to the imitation game: after the room is locked, why is C unable to distinguish whether the data is produced by a human or a machine, by A or B? I think this has to do with the mode of data production. Data, as an information product, has its value determined, as is well known, by the socially average necessary labor time (SANT) related to its production. When X is a human using real environmental monitoring facilities to produce online monitoring data, then SANT is not equal to 0. As mentioned earlier, it contains the crystallization of human undifferentiated living labor. Precisely because humans are needed to ensure the integrity and completeness of the facilities, we denote the labor time invested by humans to achieve that purpose as TREC. Due to the existence of TREC, we reaffirm that $SANT \neq 0$. In contrast, in the process where a machine outputs data, it may also seem that human labor is required to ensure the integrity and completeness of the server facilities that allow artificial intelligence to operate. However, one premise is broken: the server facilities are not the relevant facilities for the production of this type of data. In other words, in the machine's data output process, $SANT = 0$. This ultimately leads to its evil-doing and harmful actions. We can further point out that since there is no relevant human undifferentiated living labor, its variable capital v equals 0. Then, assuming a market has formed, the surplus value m is also naturally 0. If we only have constant capital c (including server facilities, regardless of their relevance) that is not zero, then what difference is there between machine-produced data and human-produced data? Could they be in a competitive relationship? Unfortunately, as we have mentioned, current machines and artificial intelligence are generally recognized not to possess autonomous consciousness, self-reproduction capabilities, or life characteristics such as emotions, desires, feelings, and fears. They cannot even understand what competition is. Therefore, there is only one answer: leaving aside interfering statements, they are also confirmed to be wasting resources and doing evil.

In the era examined by Karl Marx and Friedrich Engels, which is even more remote than Turing's time, living labor naturally referred to physical labor. After World War II, the knowledge economy began to spread. Marx guided us with the concept of undifferentiated human labor crystallized in commodities. Conversely, in the modern world of mental labor, how should we understand undifferentiated labor, or does undifferentiated human living labor exist? One might think that the products of mental labor include theories and knowledge. But do theories and knowledge, once discovered and verified by humans, immediately complete their production? I attempt to summarize the essential characteristics of living mental labor as follows: the characteristics of the labor value chain have changed; most ordinary variable capital labor appears less important; labor time, intensity, and headcount have also become things of the past; surplus industries, surplus machines, surplus

means of production, and surplus capital have emerged in large quantities; the rate of return for small and micro capital, like the wages of traditional labor, is difficult to increase; educational investment has replaced traditional labor time. However, traditional value creation theory still exists in the tertiary industry. Meanwhile, the labor prices in the primary and secondary industries are beginning to be reflected in paying for knowledge. It is inappropriate to forcibly divide the value created by knowledge into variable capital and surplus value. This can be disproved: if there were undifferentiated living labor in the field of mental labor, could I be equated with Marx? Obviously not. Therefore, mental labor is inherently differentiated; it has little to do with labor time but is positively correlated with the quantity of high-quality labor. Research facilities and educational systems have become more typical forms of constant capital. Among the entire range of labor that can be replaced by machines, perhaps only TREC can still create some surplus value. Whether derived from practice or boldly hypothesized, the outcomes of mental labor require verification and validation. We attempt to propose the following value formula for mental labor products: $F = K + S$, Where K represents the process of verifying the proposed product of hard mental labor, containing a peak of labor quantity, and its manifestation can be intellectual property; S is the part used to create traditional surplus value in the process of applying theory to practice. F is the total value, not exclusively owned by any capitalist or bourgeoisie. However, in the realm of production relations theory, this formula still requires exploration. Since exploitation refers to the unpaid appropriation of others' labor, products, or services, even without considering the cost of using intellectual property, appropriating the outcomes of mental labor can only be achieved through understanding. Learning has a cost; it requires effort and time, making unpaid appropriation basically impossible. Hence, there is no exploitation, nor surplus value m. The formula for F remains unchanged, but K does not create surplus value because S is traditional, so it is related to the number of people. When noble and patriotic scientists forgo K (For example, the Nobel Prize created conditions for Madame Curie; Deng Jiaxian, Yao Tongbin, and others firmly believed that science has no borders, but scientists have their motherland), it indeed leads to an increase in S as the number of people increases, so the total F remains unchanged. The more S is produced, the more people benefit, and the more opportunity the original creators of mental labor have to be remembered by posterity and to transmit merit to all beings. Although limited resources mean the total amount of S can still be controlled. Of course, one point has been overlooked, which would directly violate this value theory: regardless of whether K is given up, the ownership of the results of mental labor must be clear and unambiguous, otherwise it will affect the emergence of more F. Moreover, more often than not, $F = K_N + S$ ($N = 1, 2, \dots, n$), meaning that one S requires multiple mental labor outcomes. In such cases, the value of K becomes even more difficult to determine, but the significance of the formula is not diminished; at the very least, it makes communication with investors clearer.

In the software domain, K can manifest as software copyright, an open-source algorithm, an integrated framework, or a publicly available software engineering project for information collection, chat responses, ledger maintenance, consensus encryption, etc. Then the creation of S could be the operation of a specific software or the invocation of certain program modules. Traditionally, S is still related to the number of people because software also runs

on machines. Admittedly, the learning curve for some mental labor outcomes is very steep—whether because they are not frequently used, require expensive or rare facilities, or involve immature or uncommon tools, secret usage tips, or parameters. This indirectly proves the value of the knowledge economy and the absence of exploitation.

Those familiar with the market know that although K does not create surplus value, through sharing and learning, it can create "common value." In our process of building a modern socialist strong power and creating a community with a shared future for mankind, "common value" is indispensable! Investors are, in fact, all participating in market competition, and investors are also in competitive relationships with each other. Whether in primary or secondary businesses or investment practices, apart from irrational components, all are acts of market competition. Even using "pure mental labor" products as means of production, making value judgments as competitive behavior is itself unproblematic, because market competition cannot exist without value judgments. Therefore, the era when one could beat the market solely by looking at capital/goods secondary market K-lines, technical indicators, volume profiles, and trading volumes has passed. There seem to be only three exceptions: ① Some domestic or foreign K-line technical indicators are patented and not generally accessible; they might remain effective long-term. ② Some people can remain crazy and irrational for a considerable period, but the overall irrationality of the market is, of course, decreasing over time and can be neglected! ③ Returning to the imitation game: using machines and AI to observe K-lines, technical indicators, volume profiles, and trading volumes for quantitative trading—of course, they are unlikely to look only at these things.

****10. Final Words****

The world of 1950 was completely different from today. Back then, the Chinese People's Volunteer Army crossed the Yalu River with great vigor and courage to participate in the Korean War, embarking on the glorious struggle of "Resisting US Aggression, Aiding Korea, Protecting the Homeland, and Defending the Country." This strategic move aimed to consolidate the newly established regime. However, the martyrdom of several brave souls, including Wu Shi and Zhu Feng in Taiwan Province of China, along with the failed landing operation on Jinmen Island a year earlier, marked the beginning of the division across the Taiwan Strait—a situation that has caused deep regret for almost all Chinese people ever since. If philosophers have only interpreted the world in different ways, the crucial point is to change the world and make it better. Yet, understanding and knowing the world must rely on scientists, and it is scientists who must play a leading role. Some foundational theories of production relations introduced in this article are based on the principle that the nature and level of productive forces determine the nature of production relations. When the development of productive forces breaks through the existing framework of production relations, it inevitably triggers a transformation of those relations. Production relations that adapt to the development of productive forces will promote progress; those that lag behind become fetters. Production relations, in turn, have a counteractive effect on productive forces, either facilitating or hindering them. The contradiction between productive forces and production relations, as well as between the economic base and the superstructure,

constitutes the fundamental driving force of social development. We emphasize that changes in production relations must be grounded in the actual level of productive forces, and the rapid development of productive forces will, in turn, demand and guide the evolution of production relations. The new version of the Turing Test proposed in this article serves as an example. I dare to assert unequivocally that those who claim the theory of production relations only leads to "decoupling and supply chain disruption" and equate risk reduction with de-Chinization are precisely those whose improper understanding and application of the theory lie at the heart of the problem. A correct practice of the theory of production relations can, on the contrary, bring into play the theory of comparative advantage proposed by Mr. David Ricardo, thereby promoting economic development!

To be precise, can a theory ever get an opportunity to be practiced? Whether it does or not is irrelevant to the truth of the theory. From a limited time and place perspective, both types of theories may lack the chance for practice. For example, no commercial civilian nuclear power plants have ever been built in North Korea. Yet obviously, no one in North Korea doubts the correctness of the theory of nuclear fission. The same applies to the theory of productive forces, let alone the theory of production relations. Hence, it is said that the key issue for any theory remains its truth or falsehood, and practice is the sole criterion for testing truth. Limited by the Earth's finite resources, and without considering extraterrestrial economies, the immense abundance of mental labor products and the full unleashing of innovation potential may propel socialism with Chinese characteristics into a higher stage of socialism. This will lay the foundation for building a prosperous, democratic, culturally advanced, and modern socialist country where the people achieve common prosperity and material abundance!

The world may soon witness a time when robots outnumber humans. I cannot say exactly when, but one thing is clear: valuable resources should be finite, a high-quality environment should be priceless, and harmonious coexistence between humans and nature is the goal. As low-altitude economies, space exploration, and commercial aerospace industries emerge and flourish vigorously, we must first take responsibility for the Earth beneath our feet. This is necessary and unavoidable. In my middle school textbook, the famous Soviet writer Nikolai Ostrovsky left behind a passage that I can still recite fluently: " Man's dearest possession is life. It is given to him but once, and he must live it so as to feel no torturing regrets for years without purpose, never know the burning shame of a mean and petty past; so that, dying, he can say: 'I have given my whole life and all my strength to the cause of mankind—the fight for the liberation of humanity!'" My current understanding is that the liberation of humanity cannot be achieved without machines and intelligence. But the liberation of the human mind requires even better intelligence and machines. I hope the people of Ukraine have not forgotten their revolutionary forebear.

Somebody have suggested naming the new version of the Turing Test, the related paradox, the demand analysis theory, and TREC after me. Perhaps that is a good idea.

To sum up, the era of unbridled, lawless growth of machines and intelligence must come to

an end! Let us open the door to the intelligence of the real physical world and the building of an ecological civilization using high-quality, valuable big data from environmental monitoring!

For the future of the Earth and the well-being of generations to come—before the Earth is damaged and polluted beyond repair—I cannot help but loudly appeal: On this difficult journey of taking responsibility for the Earth, let all people around the world with a correct worldview, respect for science, and adherence to truth unite! Long live the great General Secretary Xi Jinping! Long live the brilliant thoughts of the great General Secretary Xi Jinping on socialism with Chinese characteristics for a new era!

We can only see the near future, but we can see that there is much work to be done there.